Bayesian Actor-Critic Wave Energy Converter Control With Modeling Errors

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***Abstract*—This paper presents a comparison of a Reinforcement-Learning (RL) based wave energy conversion controller against standard reactive damping and model predictive control (MPC) approaches, in the presence of modeling errors. Wave energy converters (WECs) are under the influence of many non-linear hydrodynamic forces, yet for ease and expediency, it is common to formulate linear WEC models and control laws. Therefore it is expected that significant modeling errors may be present, which may degrade model-based control performance. Model-free RL approaches to control may offer a significant advantage in robustness to modeling errors, in that the model is learned by the controller by experience. It is shown that, for an annual average sea state, RL-based controllers can outperform model-based control – reactive control and MPC – by 19% and 16%, respectively, when significant modeling error is present. Furthermore, compared to similar studies of RL-based control, the proposed model can reduce the training time from 8.4 hr to 1.5 hr.**

***Index Terms*—Wave energy converter, Bayesian optimization, actor-critic, model predictive control, reactive control.**

I. Introduction

OCEAN wave energy can potentially provide a great contribution to electrical energy production, with an approximated global resource of 146 TWh/yr [1]. One of the functional challenges is to maximize the power generation while preventing the wave energy converters (WECs) from mechanical failures in high-energy sea states. The control policy of a WEC plays a crucial role in maximum energy absorption from ocean waves. Many control strategies have been studied and investigated for optimal WEC operation. A review of control techniques can be found in [2], [3]. With linear operation, control techniques such as complex-conjugate control [4] can result in optimal power production of WECs, though there are barriers to applying optimal control in realistic operational conditions. Furthermore, model predictive control (MPC) is proposed to enable the controller to respond to the constraints ahead of time and apply a control force that maximizes the power production [5]–[7]. Note that for the previously mentioned techniques, controller performance depends on the accuracy of wave excitation force predictions [3] which can be a challenge.

A simplified form of complex-conjugate control, in which the PTO force is applied proportional to velocity and position, referred to as reactive control here, relies on time-averaged sea states, therefore it is required to assume wave conditions are stationary during the specified period of time [8]. To maximize the power generation in each sea state with a reactive control, simulations are run to determine the optimal Power Take-Off (PTO) parameters [9].

However, one of the challenges with many of the model-based control approaches, such as MPC, is that they often require significant domain expertise from the control engineer and any modeling errors can result in a significant control degradation. Techniques from robust control can be employed to guarantee certain limits of performance given modeling errors, but again some form of a priori model and estimates of error ranges are required.

In [10], the authors have examined the impacts of modeling and prediction errors on the performance of multi Degree-Of-Freedom (DOF) WECs. They applied an MPC law to the multi-DOF WaveSub WEC in idealized and more realistic scenarios and they showed that the model mismatch is very important, and can result in a significant reduction in power generation in comparison to the ideal scenario. The paper [11] discusses the sensitivity of controller performance to modeling errors for different WEC control system architectures and shows that it is of potentially greater impact than for traditional regulation/servomechanism control problems. A hierarchical control technique for improving the energy absorption in heaving WEC is presented in [12]. Results show the effectiveness of their proposed higher and lower level controller in the nominal case as well as in the presence of parametric uncertainties, external disturbance, and unmodeled forces.

To deal with modeling errors, model-free Reinforcement Learning (RL) techniques are introduced in [13], [14] with a discrete state space and discrete action space. However, a continuous control space could be more effective for WECs by avoiding quantized discontinuities in the control action. Deep RL algorithms such as Q-Actor-Critic (QAC) and gradient methods can be utilized to implement continuous controllers [9], [15]–[17]. Monte-Carlo techniques are deployed in many policy gradients to calculate the gradient that is used to adapt a parameterized policy. Monte-Carlo techniques tend to have high variance resulting in a slow convergence [18]. In this paper,

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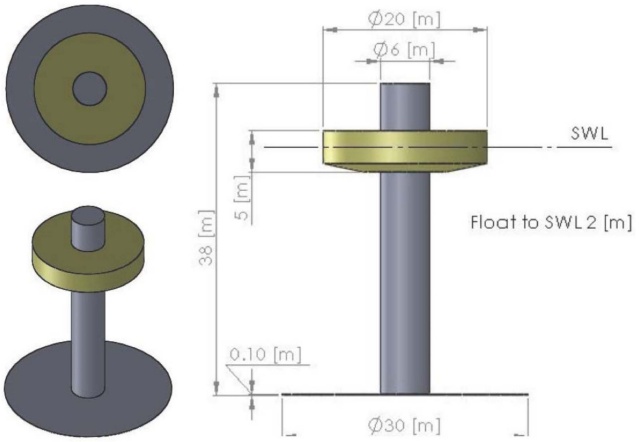


Fig. 1. Dimensions of the wave energy converter reference model RM3.

we propose a Bayesian Actor-Critic (BAC) scheme for a WEC control and power optimization. Bayesian policy gradient helps decrease the data samples required to compute accurate gradient estimates. We supplemented the Bayesian policy gradient with Actor-Critic to take advantage of the Markov property for fully or partially observable Markov Decision Processes (MDPs). RL algorithms are computationally more complex than other model-based WEC controllers. However, they are more robust to modeling errors and non-linear models. In this study, we examined and compared our proposed BAC model performance with standard MPC and reactive damping control in the presence of modeling errors. The results show that the proposed BAC model is more robust to modeling errors compared to MPC and reactive control. We evaluated model-based and model-free methods performance in terms of mean absorbed power, the mechanical power at the interface to the PTO. Future work could include a consideration of electrical power and generator losses.

II. WEC Model Hydrodynamics

A. RM3 Reference WEC Model

In this study, a two-body point absorber – the Reference Model 3 (RM3) – [19] is used for the proposed Bayesian Actor-Critic model performance evaluation. The two-body point absorber is simulated in the open-source Wave Energy Converter Simulator (WEC-Sim) – developed by Sandia National Labs and the National Renewable Energy Lab [20]. RM3 [21] consists of two bodies, a float and a plate connected by a PTO device. Fig. 1 shows the device specification. Power is captured mainly in the relative heave direction, although the WEC is free to move in all six degrees of freedom in regards to wave motion. RM3 operation is relatively straightforward and representative of a general point absorber with a float and a reaction plate. Fig. 1 shows the geometry and dimensions of the testing model, and its mass properties are listed in Table I.

WEC-Sim is a radiation and diffraction based numerical model that was designed to solve the system dynamics of multibody wave energy converters, PTO systems, and mooring systems [22]. The dynamic response in WEC-Sim is computed

TABLE I
Mass Properties for RM3 Two-Body Point Absorber

Body	Mass (kg)	Center of Gravity (m)	Diagonal Moment of Inertia at Cg (kg m^2)
Float	3.562e6	[0 0 -0.72]	[4.09e7 4.14e7 3.71e7]
Plate	2.172e7	[0 0 -21.29]	[4.65e8 4.64e8 0.28e8]

by solving the motion equation of the float and plate, based on the Cummins equation, shown below for a single body:

$$(m + A_\infty)\ddot{x} = F_r + F_e + F_{vd} + F_{res} + F_{pto} + F_{mo} \quad (1)$$

where A_∞ is the added mass matrix at infinite frequency, x is the 6 DOF displacement, m is the mass matrix, and F_r, F_e, F_{pto}, F_{mo}, F_{vd}, and F_{res} are the radiation force, wave excitation force, PTO force, mooring force, quadratic viscous drag, and net buoyancy restoring force matrices for all coupled 6 DOF [22]. In this research, we simplify the system to two bodies, each operating in the single heave degree of freedom, z_1 and z_2, with the PTO force and drag force reacting between the bodies.

$$(m_1 + A_{\infty,1})\ddot{z}_1 = F_{r,1} + F_{e,1} + F_{vd} + F_{res,1} + F_{pto} + F_{mo,1} \quad (2)$$

$$(m_2 + A_{\infty,2})\ddot{z}_2 = F_{r,2} + F_{e,2} - F_{vd} + F_{res,2} - F_{pto} + F_{mo,2} \quad (3)$$

$$z = z_1 - z_2 \quad (4)$$

The PTO force is the control parameter and we deployed a model-free RL based model to tune PTO force in order to maximize the mean absorbed power. The absorbed PTO power is defined as

$$P_{pto} = -F_{pto}\dot{z} \quad (5)$$

and the mean absorbed power is

$$P_{pto,mean} = \frac{1}{T}\int_0^T -F_{pto}(\tau)\dot{z}(\tau)\, d\tau \quad (6)$$

and T is the simulation length.

B. Viscous Drag and Modeling Error

To create a modeling error, we have added a quadratic viscous drag term to the plant model, that is not included in the control models.

$$F_{vd} = k_{vd} \cdot (-1 \cdot 10^6) \cdot |\dot{z}|\dot{z} \quad (7)$$

Where k_{vd} is reflective of modeling error degree and $\dot{z}$ is the relative heave velocity between float and plate. This drag term could be utilized to model non-linear effects not included in the original boundary element method solutions, or marine growth on the structure over time. The viscous drag term will introduce a modeling error, but it will also decrease the power conversion capability of the WEC. Therefore, as the tuning parameter k_{vd} is increased, we expect to see controller performance degradation due to both the increase in unmodeled behavior and also the increased power loss due to extra damping.

To relate the k_{vd} tuning parameter to modeling error, we first define a base damping of 2.89e6 N/(m/s), based on the RM3

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hydrodynamics around typical wave periods. Then, linearizing (7) around an average absolute speed of $\dot{z} = 0.16$ m/s, as is observed over typical sea states, we define the modeling error as

$$\text{modeling error pct} = \frac{k_{vd} \cdot (1e6) \cdot 2(0.15)}{2.89e6} \cdot 100 \quad (8)$$

$$= 11.1 \cdot k_{vd} \quad (9)$$

Therefore, as a rough approximation, each integer increase in k_{vd} corresponds to an approximate increase in 10% of modeling error.

III. WEC Control

A. Reactive Control

A standard WEC reactive control policy is used as a performance benchmark in this study. The reactive control law with two control parameters is formulated as follows:

$$F_{pto} = B_{pto}\dot{z} + C_{pto}z \quad (10)$$

Where F_{pto}, B_{pto} and C_{pto} represent the PTO force, PTO damping, and PTO stiffness, respectively, and $\dot{z}$ and z are the relative heave velocity and displacement between float and plate [9].

The optimal PTO coefficients are determined to optimized power conversion, and are based on the wave energy period, T_e, and the significant wave height, H_s, of the sea state. (Consideration of maximum excursions in velocity or displacement to respect system physical constraints could also be considered in the selection of the reactive control parameters.) Under greatly simplifying conditions, such as an assumption of regular seas, the optimal damping and stiffness can be determined analytically, but in irregular sea conditions, and especially if consideration of system limits on displacement and velocity are given, the damping and stiffness for a given sea state are determined experimentally.

B. Model Predictive Control

Model Predictive Control is a technique in which the current control action is defined by solving a finite-horizon control problem at each sampling time using the most recent state of the plant as the initial step and solving for an optimal control sequence [23], [24]. This process is repeated every time step. The constrained look ahead online optimization capability of MPC makes it a strong candidate for many industrial systems and processes. MPC is a very encouraging control method for WECs due to the relatively slow plant dynamics because it optimizes power generation while observing the system's mechanical limits [5]. Since the MPC algorithm uses an internal plant model to predict the system's future states, the analysis of the stability, robustness, and sensitivity of MPC is more challenging to handle than linear feedback controllers [25]. Demand for a very accurate WEC model may create a need to incorporate nonlinearities, such as viscous drag effects and mooring forces in the WEC dynamic plant model, which results in a nonlinear plant model. A Nonlinear Model Predictive Controller (NMPC) problem for WECs with nonlinear mooring forces has been addressed in [26], and nonlinear viscous drag effects have been considered in [27]. Moreover, the cost function to be optimized can take on a standard quadratic form, or it can be a non-standard higher-order piecewise functional [27] of design variables. It can be a challenge to formulate an appropriate cost function for the MPC, and constraint optimal control capability of MPC can be linked to existing linear control techniques, for example, observer-based MPC [28], state-feedback or linear quadratic regulator form [29], and generalized inverse matching [30]. The main shortcomings of MPC-based control are the computational complexity of the constraint online optimization algorithm, especially for real-time applications, and a need to develop an accurate plant model for state predictions.

The Model Predictive Control scheme used in this research is shown in MATLAB/Simulink in Fig. 2.

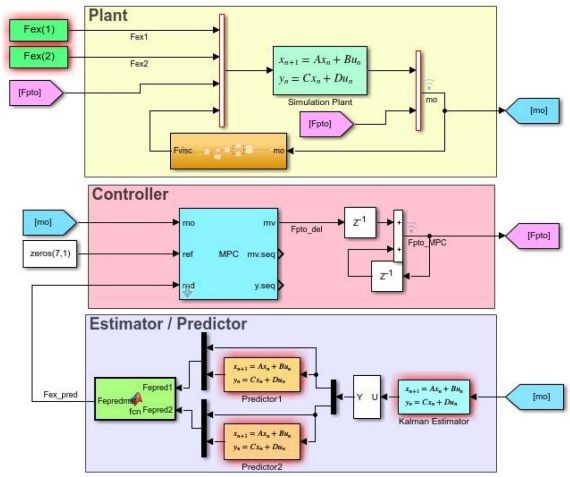


Fig. 2. MPC scheme in MATLAB/Simulink/WEC-Sim with the RM3 reference WEC.

Algorithm 1: Bayesian Actor-Critic Algorithm.

1: **algorithm initialization:**
2: initial policy parameters: $\theta \sim N(0, 1)$
3: L>0 iterations for gradient evaluation
4: $\epsilon > 0$ termination condition
5: done=false
6: **while** not done:
7: run **Gaussian Process Temporal Differences(GPTD)**
8: for L iterations to get:
9: α_t, C_t, U_t, G_t matrices
10: calculate: $\triangle\theta = U_t\alpha_t$
11: update policy: $\theta = \theta + \beta \triangle \theta$
12: **if** $\| \triangle \theta\| \leq \epsilon$ **then** done=true
13: end **while**
14: **Retrun** updated policy parameters θ^*

C. Bayesian Actor-Critic Control

Reinforcement Learning is a class of learning problems in which an agent interacts with an unknown, dynamic and stochastic environment, where the agent's objective is to adapt a policy to maximize the total reward it receives over the long run. In Actor-Critic (AC) methods, policies are parameterized

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functions and can be defined as $\{\mu(.|x;\theta), x \in X, \theta \in \Theta\}$. The gradients of the expected return, $\eta(\theta) = E[\sum_{t=1}^{T} \gamma^t R(x_t, a_t)]$, are calculated with regard to the policy parameters θ from data trajectories [31]. Afterwards, the policy parameters are updated in the direction of the gradient accordingly. Based on the policy gradient proposition [32], the gradient of the expected return can be computed by

$$\bigtriangledown_\theta \eta(\theta) = \int_z dz \pi(z;\theta) \bigtriangledown_\theta log(\mu(a|x;\theta)) Q(z) \tag{11}$$

where $\pi(z;\theta)$ is a discounted weighting of state-action pairs and $Q(z)$ represents the expected cumulative return. The BAC method is based on a statistical generative model that relates the observed reward signal $R(z)$ to the unobserved action-value function $Q(z)$ as

$$R(z_i) = Q(z_i) - \gamma Q(z_{i+1}) + N(z_i, z_{i+1}) \tag{12}$$

where $N(z_i, z_{i+1})$ is a zero-mean noise signal that represents the discrepancy between $R(z_i)$ and $Q(z_i) - \gamma Q(z_{i+1})$. The finite-dimensional processes $Q(t)$, $N(t)$, $R(t)$, and the $t \times (t+1)$ matrix $H(t)$ are defined as follows:

$$\begin{aligned} &R_t = (R(z_0), \ldots, R(z_t))^\mathsf{T}, Q_t = (Q(z_0), \ldots, Q(z_t))^\mathsf{T} \\ &N_t = (N(z_0, z_1), \ldots, N(z_{t-1}, z_t))^\mathsf{T}, \end{aligned} \tag{13}$$

$$H = \begin{bmatrix} 1 & -\gamma & 0 & \ldots & 0 \\ 0 & 1 & -\gamma & \ldots & 0 \\ . & & & & . \\ . & & & & . \\ 0 & 0 & 0 & 1 & -\gamma \end{bmatrix} \tag{14}$$

The set of (12) for $i = 0, ..., t-1$ can be represented as $R_{t-1} = H_t Q_t + N_t$. Under certain assumptions on the distribution of the discounted return random process [33], the covariance of the noise vector N_t is given by

$$\Sigma_t = \sigma^2 H_t H_t^T = \sigma^2 \begin{bmatrix} 1+\gamma^2 & -\gamma & 0 & \ldots & 0 \\ -\gamma & 1+\gamma^2 & -\gamma & \ldots & 0 \\ . & & & & . \\ . & & & & . \\ 0 & 0 & \ldots & -\gamma & 1+\gamma^2 \end{bmatrix} \tag{15}$$

The Bayesian AC algorithm is composed of two components: a non-parametric Bayesian Critic and a parametric Bayesian Actor. The Critic model determines the posterior distribution for the state-action value function and the Actor model uses the posterior distribution computed by the Critic and calculates the posterior distribution with policy gradients [33]. The posterior mean represents the estimate of the policy gradient and is utilized for the policy function update [15]. On the other hand, the posterior covariance is an illustration of the update reliability. In the next section, we explain the formulation of the BAC algorithm followed by the proposed controller scheme.

Bayesian Actor: The Bayesian Actor is to compute the posterior distribution for the policy function gradient given the trajectory data D_t. The policy gradient can be expressed as in (11). A Gaussian Process prior is placed over the state-action value function using a prior covariance kernel function $k(z, \acute{z}) = Cov[Q(z), Q(\acute{z})]$. Since the expected return (11) is linear in Q and g can be denoted as $g(z;\theta) = \pi(z;\theta) \bigtriangledown_\theta log(\mu(a|x;\theta))$, therefore the posteriors of the policy gradient can be calculated as follows:

$$E[\bigtriangledown_\theta \eta | D_t] = \int_Z dz g(z;\theta) E[Q(z)|D_t] \tag{16}$$

$$Cov[\bigtriangledown_\theta \eta | D_t] = \int_{Z^2} dz d\acute{z} g(z;\theta) Cov[Q(z), Q(\acute{z})|D_t] g(\acute{z};\theta)^T \tag{17}$$

Bayesian Critic: The Bayesian Critic is structured to supply the Actor with the posterior substances of $Q(z)$. The Gaussian Process Temporal Difference (GPTD) provides a well-established process for this step [33]. The GPTD model is a direct application of Gaussian Process Regression (GPR) in Temporal Difference (TD) learning. This is a model that correlates the observed reward instant to the unseen action-value function. By putting a Gaussian Process prior on $Q(z)$ and assuming that the noise is also normally distributed, the Bayes rule can be used to obtain the posteriors of Q as

$$E[Q(z)|D_t] = k_t(z)^T \alpha_t \tag{18}$$

$$Cov[Q(z), Q(\acute{z})|D_t] = k(z, \acute{z}) - K_t(z)^T C_t K_t(\acute{z}) \tag{19}$$

Now by replacing the expressions for the posterior substances of the state-action value function in (16) and (17) with the results of the Critic model in (18) and (19), we come by

$$E[\bigtriangledown_\theta \eta | D_t] = \int_Z dz g(z;\theta) k_t(z)^T \alpha_t \tag{20}$$

$$Cov[\bigtriangledown_\theta \eta | D_t] = \int_{Z^2} dz d\acute{z} g(z;\theta) k(z, \acute{z}) - K_t(z)^T C_t K_t(\acute{z}) g(\acute{z};\theta)^T \tag{21}$$

Equations (20) and (21) represent the general form of the posterior policy gradient, although it could be computationally challenging. By selecting the prior covariance as the summation of an arbitrary state-kernel and the Fisher kernel over state-action tuples [34], the integrals can be calculated as follows:

$$k(z, \acute{z}) = k_x(x, \acute{x}) + K_F(z, \acute{z}) \tag{22}$$

$$K_F(z, \acute{z}) = u(z)^T G^{-1} u(\acute{z}) \tag{23}$$

Where K_x is an arbitrary positive definite kernel function and K_F is the Fisher kernel in which

$$u(z) = \bigtriangledown_\theta log(\mu(a|x;\theta)) \tag{24}$$

$$G = E[u(z)u(z)^T] \tag{25}$$

are the score vector and the Fisher information matrix correlating to the policy $\mu(\theta)$, respectively. Then there is,

$$E[\bigtriangledown_\theta \eta | D_t] = U_t \alpha_t, U_t = [u(z_0), u(z_1), \ldots, u(z_t)]$$

$$\alpha_t = H_t^T (H_t K_t H_t^T + \Sigma_t)^{-1} r_{t-1} \tag{26}$$

$$Cov[\bigtriangledown_\theta \eta | D_t] = G - U_t C_t U_t^T$$

$$C_t = H_t^T (H_t K_t H_t^T + \Sigma_t)^{-1} H_t \tag{27}$$

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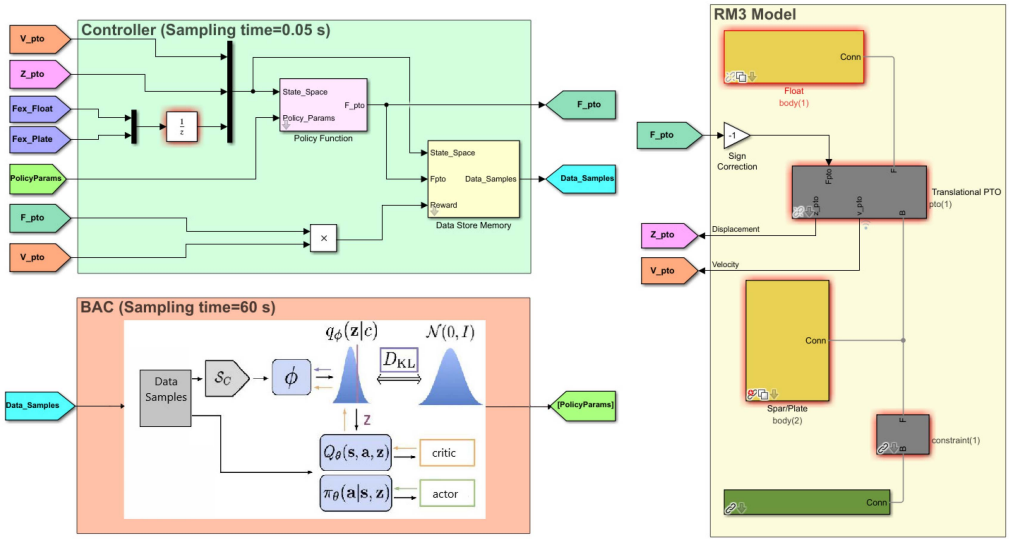


Fig. 3. BAC controller scheme.

Evaluating the Fisher information matrix G could be challenging since it requires computing the expectation over the state-occupancy density $\pi(x)$ besides the expectation with respect to the policy. However, one simple way is to estimate G from a trajectory data D_t using an unbiased estimator [15]. Algorithm 1 is a pseudocode sketch of the Bayesian Actor-Critic model, using regular gradient in the policy update. The proposed BAC control scheme used in this research is shown in MATLAB/Simulink in Fig. 3.

D. Problem Formulation

RL methods are model-free decision-making schemes that are normally implemented to train a system to do a task that might be challenging to represent in a standard control setting. These assignments are represented as episodic, i.e., the experience can be formed of multiple steps with a termination that is defined by either success or failure in reaching a predefined goal. Reformulations of the RL strategy with a continuous WEC control are studied in [14], [16].

In the BAC control design, state space and action space for an RM3 point absorber are defined as

$$\text{state} = [\hat{F}_{exf}, \hat{F}_{exp}, z, \dot{z}], \quad \text{action} = F_{pto} \tag{28}$$

Where $\hat{F}_{exf}, \hat{F}_{exp}, z, \dot{z}$ are float and plate excitation forces and the relative displacement and relative velocity between two bodies, respectively. Inputs to the WEC plant are float excitation force, plate excitation force, PTO force, and viscous force.

Defining a proper reward function is very important in RL methods to have the agent learn the optimal policy while taking the device constraints into account. In RL settings, the objective function is to maximizing the reward function instead of minimizing the cost function. Taking inspiration from [16], the reward function can be defined as

$$\text{reward} = -F_{pto}\dot{z} - W_z X_{threshold} - F_{pto}^2 W_f \tag{29}$$

where the weights W_z and W_f are the parameters used to adjust the penalty on the control action and RM3 displacement, respectively. The displacement limit of the RM3 is 5 m and $X_{threshold}$ is equal to 1 when the relative displacement exceeds its limit and zero otherwise. We set W_z and W_f to $2 \cdot 10^5$ and $7 \cdot 10^{-8}$ respectively. W_z and W_f are chosen to place a negative effect on the reward in case of exceeding the limits while maintaining the first term the dominant part of the reward function. The values for W_z and W_f were tuned based on observations of system behavior and speed of convergence. After around three episodes of training, the control model learns displacement constraints and does not exceed its limit; therefore the second term is zero after a few episodes of training. On the other hand, the third term is very important for the PTO force magnitude as well as its smoothness. A value of $7 \cdot 10^{-8}$ is selected for W_f since values lower than $5 \cdot 10^{-8}$ were not large enough to optimize the magnitude of PTO force signal especially for high-energy sea states. On the other hand values higher than $9 \cdot 10^{-8}$ for W_f were too large for low energy sea states pushing PTO force signal to go lower than its optimal range. We decided to use $7 \cdot 10^{-8}$ which is the mid-point of this range.

The other difference between the RL-based and MPC schemes is the prediction phase of the upcoming waves. In MPC, an external system [35] is implemented to predict the future wave force and the data is embedded in the cost function to choose the proper control input. In the RL setting, the goal is to have the agent learn about the environment and train an optimal policy function to maximize the total reward which total reward is the summation of the instant reward and discounted future rewards. Thus, the reward function needs to be determined based on the current step rather than including data from upcoming time horizons. The prediction phase in the RL system is designed with a probabilistic setting [16].

IV. Bayesian Actor-Critic Model Simulation Results

A. Case Study

The RM3 reference model point absorber, developed by Sandia National Laboratories, is selected as a case study for the development and evaluation of the proposed Bayesian

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TABLE II
SEA STATES CHARACTERISTICS

Sea State	Significant Height (m)	Energy Period (s)
Winter	3.3	10.5
Annual Average	2.4	9.0
Summer	1.6	7.3

Actor-Critic control scheme. MATLAB/Simulink/WEC-Sim is used to simulate the plant and controllers.

B. Water Surface Elevation Data

In order to have enough data samples for the training of the BAC control model, 10 time series of irregular waves with 20 minutes time duration are generated, with the significant wave height ranging from 1.0 m to 3.5 m and the energy period from 7.0 s to 11.0 s. A Bretschneider spectrum is used to generate water surface elevation data with a sampling time of 0.05 seconds. Time-series with different random phase seeds are created for training and evaluation of the control schemes. Three sea states including winter sea state, annual average sea state, and summer sea state are chosen for control models evaluation. The selected sea states are the typical sea states over a full year, summer months, and winter months at station 46050 of the US Pacific Northwest [36]. Details of these three sea states can be found in Table II.

C. Simulation Results

Training of the BAC model starts without any prior knowledge and the policy takes random actions in the early stage of learning. Each episode contains a randomly initialized wave series whose significant wave height and energy period are randomly chosen in the 1.0 to 3.5 m and 8 to 11 s range, respectively. The wave trace for each episode lasts for 60 s and the BAC model is updated at the end of each episode. Since the control time step is 0.05 s, each episode contains 1200 samples for which the first 200 samples are not used for training to avoid numerical unreliability. A training length of 60 seconds (i.e., 1200 samples) was found to strike a good balance between the amount of data and the training convergence speed. As seen in Fig. 4, the agent learns the nearly optimal policy to maximize the expected reward after 30 episodes. After 90 episodes the BAC policy shows a close performance to the optimized reactive control model. Ninety episodes corresponds to an approximately 1.5 hr time series. This is a very short time series required for training WEC control systems and provides confidence in the controller for real-world implementations. The overall reward during an episode depends on the energy level of the waves. Thus, even after convergence, there are potentially large variations in the total reward per episode. Due to these large variations, the mean, minimum and maximum reward of ten iterations of training are considered for the BAC controller performance evaluation. Combining Bayesian with Actor-Critic improves the learning process significantly. In [16], the authors used the standard Actor-Critic as the WEC controller, on a one-body point absorber; the convergence time for their controller was about

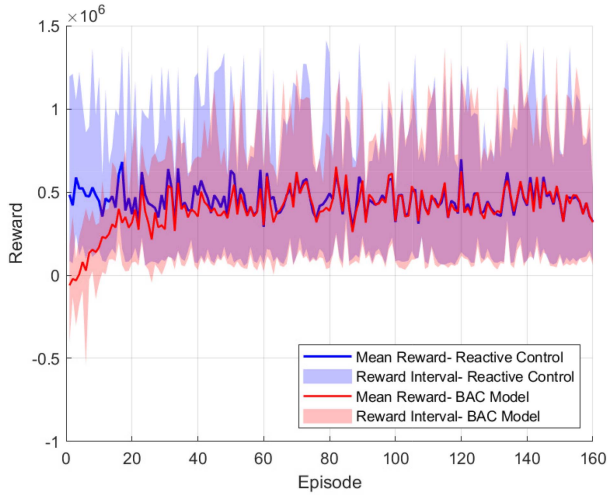


Fig. 4. Mean reward per episode during BAC model training. Convergence occurs by episode 90.

8.4 hr. In this study, With Bayesian Actor-Critic this convergence time drops to 1.5 hr which proves the proposed model's data efficiency. Furthermore, the convergence plot shows that the BAC model is more stable compared to standard Actor-Critic since it does not suffer from over-training and does not show aggressive behavior as the standard Actor-Critic had shown in [16]. The proposed control scheme is compared against the standard reactive control assuming there are no modeling errors considered in the plant definition. There is no training process for the reactive control model since the optimal control parameters are pre-calculated and given to the RC model, so it can be used as a benchmark for BAC model evaluation. As shown in Fig. 4, after convergence the BAC control model shows a close performance to the optimized reactive control model.

In this study, MPC and reactive control are chosen as benchmark schemes to evaluate BAC model performance. MPC and reactive control models are model-based methods; therefore, their operation depends on the plant definition. Modeling errors, which can come from unmodeled high-order drag effects, or changes to the plant over time, such as marine growth on the structure, can cause suboptimal control, thus reducing power capture [9]. Any modeling errors affect model-based controllers training as well as their performance. On the other hand, RL approaches are model-free techniques, thus they can be more robust to plant modeling errors. In this study, we added the viscous damping force as the fourth input to the motion equation and as a modeling error factor to examine the MPC, Reactive Control, and the BAC performance in the presence of modeling errors. Fig. 5 shows the three introduced control models' performance in terms of mean absorbed power with respect to different percentages of modeling errors for Winter, Annual Average, and Summer sea states. The BAC model outperforms MPC and reactive control model as modeling error increases because it is a model-free method and can adjust its policy better with any introduced uncertainty and incompleteness of the plant design. Table III presents the mean absorbed power of the three control

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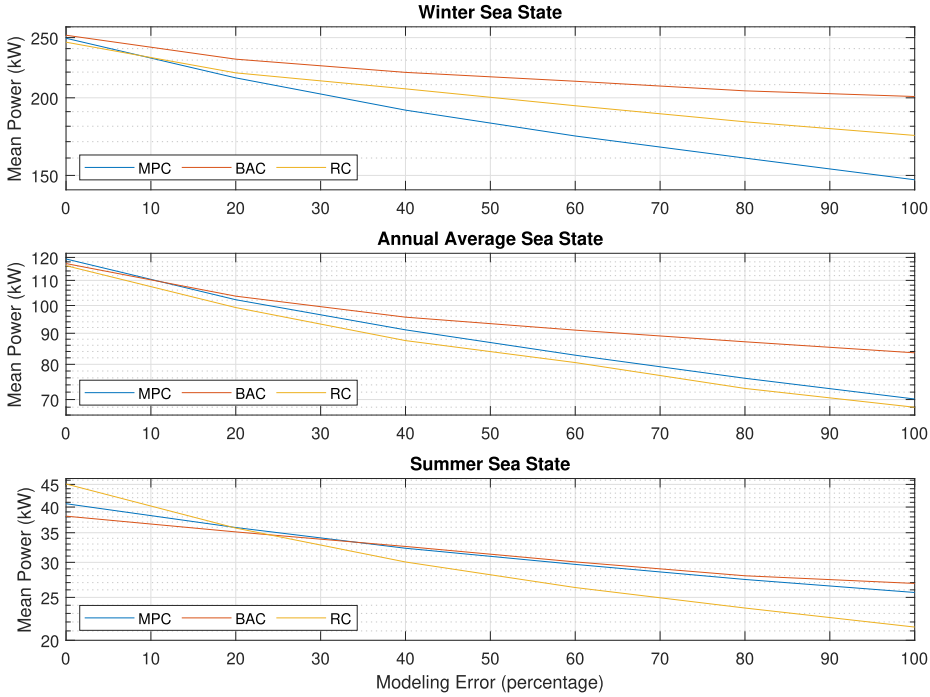


Fig. 5. RC (Reactive Control), MPC (Model Predictive Control), and BAC (Bayesian Actor-Critic) performance over Winter, Annual Average, and Summer sea states, as a function of modeling error. Note that performance degrades in all cases of increasing viscous damping (i.e., modeling error) as expected, but the performance degradation is less with the BAC reinforcement learning control, as it is model-free.

TABLE III
MEAN POWER [kW] USING MPC, RC AND BAC CONTROL IN THE PRESENCE OF MODELING ERROR

Sea State	Controller	Mean Power [kW]					
		Modeling Error Percentage					
		0%	20%	40%	60%	80%	100%
Winter	RC	245.80 (-3%)	219.37 (-5%)	206.75 (-6%)	194.20 (-9%)	182.98 (-11%)	173.97 (-13%)
	MPC	249.40 (-1%)	215.33 (-7%)	191.06 (-13%)	173.60 (-19%)	159.95 (-22%)	147.59 (-27%)
	BAC	252.20	230.79	219.76	212.67	205.23	200.95
Annual Ave.	RC	116.30 (-1%)	99.22 (-4%)	87.54 (-9%)	80.54 (-12%)	73.01 (-17%)	67.98 (-19%)
	MPC	119.30 (+2%)	102.21 (-2%)	91.18 (-5%)	82.82 (-9%)	75.88 (-13%)	70.14 (-16%)
	BAC	117.25	103.61	95.67	91.09	87.14	83.57
Summer	RC	45.10 (+18%)	35.85 (+2%)	30.06 (-8%)	26.32 (-12%)	23.65 (-20%)	21.42 (-20%)
	MPC	40.70 (+7%)	35.96 (+2%)	32.28 (-1%)	29.68 (-1%)	27.44 (-2%)	25.65 (-5%)
	BAC	38.15	35.15	32.59	30.07	27.98	26.89

models over the three sea states with different levels of modeling errors. The trailing number in parenthesis is the change from the BAC baseline, showing a decrease in power for the model-based control methods relative to the RL based BAC method. It is shown that in all the three sea states, the BAC model outperforms MPC and RC model in the presence of significant modeling errors. Assuming that there is 100% modeling error, the BAC model outperforms MPC by 27% in the winter and RC by 20% in the summer. It should be noted that in the summer sea state when there is low modeling-error, MPC and RC generate more energy than the BAC model. However, the summer sea state is low energy, so the difference in power at the low modeling-error condition is quite small.

In a control system design, the cost function depends on many parameters. In wave energy converter design, PTO force level and relative displacement between float and plate are directly related to the capacity, size, and overall capital cost of the system. In Table IV, the third standard deviation for PTO force and relative displacement on the three sea states are given, and as it can be seen BAC model has the highest PTO force and displacement compared to the other two methods. This implies that deploying a BAC controller will be more costly than the

TABLE IV
VARIABILITY IN PTO FORCE AND VELOCITY, AS 3RD STANDARD DEVIATION

Sea State	Controller	F_{pto} 3σ [N]	Z_{pto} 3σ [m]
Winter	RC	3.55e6	0.99
	MPC	2.97e6	1.15
	BAC	3.87e6	1.59
Annual Ave.	RC	1.85e6	0.78
	MPC	2.02e6	0.74
	BAC	1.75e6	1.45
Summer	RC	8.42e5	0.55
	MPC	1.13e6	0.38
	BAC	1.15e6	0.6

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TABLE V
OPTIMAL REACTIVE CONTROL PARAMETERS

Sea State	Damping [N/(m/s)]	Stiffness [N/m]
Winter	5.6e6	-2.5e5
Annual Ave.	3.2e6	-2.5e5
Summer	1.7e6	-2.5e5

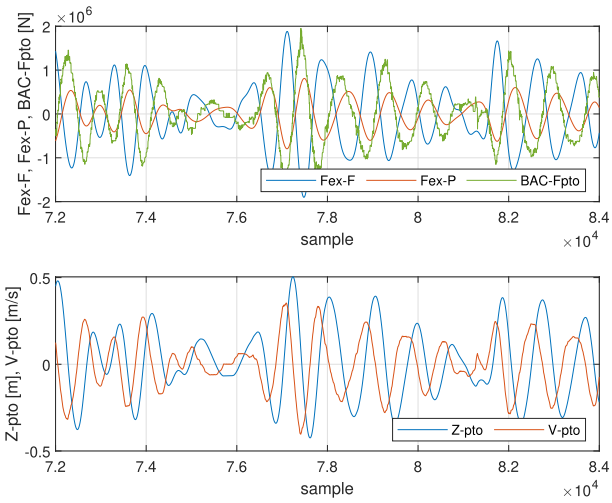


Fig. 6. Time variation of the float excitation force, plate excitation force, PTO force (top) and relative velocity and relative displacement (bottom) for the simulation of the WEC in an irregular sea with $H_s = 2.4$ m and $T_e = 9$ s.

MPC and RC methods, although the difference is not significant. Maximum force and displacement could be explicitly included in the BAC cost function if desired, in which case they could be part of the optimization.

Table V presents the optimal reactive control parameters for the three sea states. According to the data, as the energy level of a sea state increases, the optimal damping tends to go higher.

An example time-series of the system performance under BAC control is shown in Fig. 6. It can be seen that there is a high-frequency component on the PTO force command signal. Depending on the type of PTO, this high-frequency component may be undesirable. It could be reduced or eliminated either through the inclusion of a PTO rate-of-change penalty in the cost function or low-pass filtering at the controller output.

V. DISCUSSION AND CONCLUSION

In this paper, we present a control scheme based on Bayesian optimization and Actor-Critic architecture, applied to a two-body point absorber. For parametric policies, by using Gaussian Processes and the prior distributions, a posterior distribution of the policy gradient updates can be obtained. The resulting algorithm uses the Markov property of the system dynamics (if the system dynamic is Markovian) and this significantly accelerates the training process and model convergence. Unlike model-based methods, model-free RL control approaches may offer a significant advantage in robustness to modeling errors, in that the model is learned by the controller by experience.

We examined the benefits of Gaussian Processes in the control model's data efficiency and its robustness in the presence of modeling errors. Water surface elevation data of three sea states from a station in the US northwest proved the proposed model's data efficiency and its robustness against modeling errors compared to other standard WEC control models. Results show that, in the presence of significant modeling error, RL-based controllers can outperform model-based control – reactive control and MPC – up to 20% and 27%, respectively. In addition, compared to similar studies, the proposed model reduced the training time from 8.4 hr to 1.5 hr.

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